UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Saturday 8 June 2019 13:30 to 16:30

Paper PHA/WP

PHARMACOLOGY (Written Practical)

*Answer **all three** questions.*

*Start **each** question on a new sheet*

*Tie the answer to each question separately, with your **candidate number** on each cover sheet*

Calculator – students are permitted to bring an approved calculator

STATIONARY REQUIREMENTS

Metric graph paper x 5 sheets
4-cycle semi-log graph paper x 4 sheets
Script paper
Cover sheets

SPECIAL REQUIREMENTS

Approved calculator

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the invigilator

Question 1

The apelin receptor is a G-protein-coupled receptor (GPCR) that is activated by the apelin family of peptides. The apelin receptor and apelin peptides have roles in both embryonic development of the heart and the adult cardiovascular system. For example, pulmonary arterial hypertension (PAH) is a condition in which pulmonary vascular remodelling causes right ventricular failure and death. In PAH decreased apelin levels have been observed, such that there is great interest in enhancing apelin signalling as a new therapeutic avenue in PAH.

The apelin family of peptides includes different versions of apelin owing to its cleavage, the most abundant form in humans is a 13-amino acid peptide where the first amino acid is pyroglutamate, [Pyr^1]apelin-13 (N-termini pyroglutamates protect peptides from exopeptidase degradation). A different gene encodes the elabela (ELA) peptide, which like apelin can be cleaved to different lengths to produce ELA-32, ELA-21 and ELA-11. Studies in zebrafish have proposed that ELA peptides also activate the apelin receptor, but the role of ELA peptides in the human cardiovascular system is unknown. The amino acid sequences of [Pyr^1]apelin-13, ELA-32, ELA-21 and ELA-11 are given below.

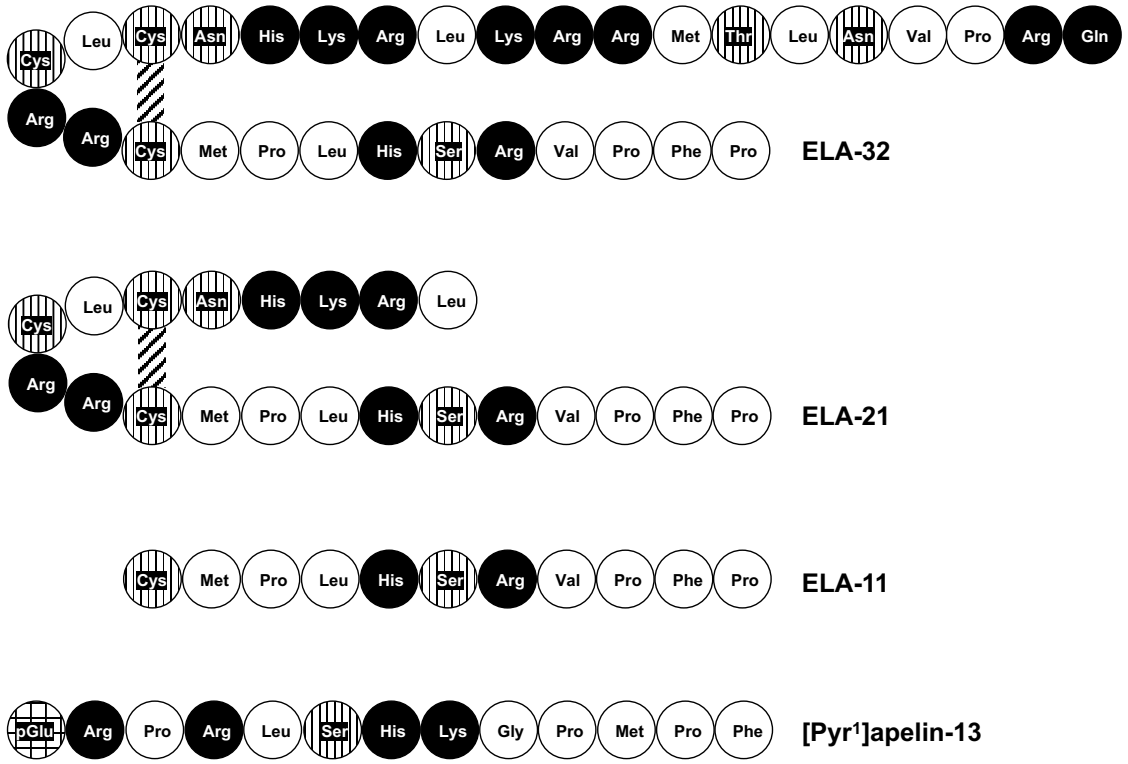

Figure 1. *The amino acid structures of [Pyr^1]apelin-13, ELA-32, ELA-21 and ELA-11. Disulphide bridges between cysteines in ELA-32 and ELA-21 are shown as diagonally hashed lines, hydrophobic amino acids are shown in white (e.g. Pro, proline), uncharged amino acids with vertical lines (e.g. Ser, serine), basic amino acids in black (e.g. Lys, lysine) and pyroglutamate with a checked pattern (pGlu).*

To determine the ability of [Pyr^1]apelin-13, ELA-32 and ELA-11 to bind to the human apelin receptor, a homogenate of human ventricles was prepared and the binding of [Pyr^1]apelin-13 and ELA-11 was measured in the presence of 10 nM of the apelin receptor antagonist [^{125}I]repulsin, which has an affinity constant of 1.3×10^9 M^{-1} for the apelin receptor. The results obtained are given in Table 1.

[Question 1 continues on the NEXT PAGE]

Table 1: *Binding of apelin receptor ligands to human ventricles*

	Specific binding (%)	
[ligand] nM	[Pyr^1]apelin-13	ELA-11
0.1	100	100
0.3	93	98
1	61	92
3	30	83
10	9	61
30	3	34
100	0	12
300	0	4
1000	0	0

A) What is the affinity constant of [Pyr^1]apelin-13 and ELA-11 for the apelin receptor? (6 marks)

B) Using the same experimental set up as described for the data in Table 1, the affinity constant of ELA-32 was calculated as 3.2 x 10^{10} M^{-1}. What might explain the basis of the different affinity constants? (1 mark)

In a second set of experiments, the receptor density was measured in homogenates of ventricles obtained from healthy human hearts and hearts from humans with PAH. The maximal binding was observed to be 4.24 fmol/mg for healthy ventricular homogenates and 3.50 fmol/mg for PAH ventricular homogenates; femto, 10^{-15}.

C) How many receptors are there in each gram of the different ventricular tissues, assuming that membrane protein is 2.7% of wet weight of this tissue? What can you can conclude about apelin receptors in PAH? Avogadro's number is 6.022 x 10^{23} mol^{-1}. (3 marks)

Next, experiments were conducted to look at different G protein signalling pathways for different apelin receptor ligands when binding to the apelin receptor.

In a cell line expressing the apelin receptor, increases in cyclic adenosine monophosphate (cAMP) concentration following the application of the adenylate cyclase activator forskolin were measured. Next, the effect of ELA-11, ELA-21 and ELA-32 on the forskolin-induced increase in cAMP levels was measured and the potency of peptides to inhibit cAMP production was found to be highly similar. In contrast, when examining the potency of all 3 peptides to recruit β-arrestin, ELA-11 had a significantly lower potency than both ELA-21 and ELA-32.

D) What can you conclude from these experiments? (3 marks)

[Question 1 continues on the NEXT PAGE]

Next, the ability of [Pyr^1]apelin-13 to recruit β-arrestin was determined in the presence and absence of various concentrations of the novel apelin receptor antagonist pogonalin. The [Pyr^1]apelin-13 EC_{50} for β-arrestin recruitment in the absence of pogonalin was found to be 1.65 nM. The results in the presence of different concentrations of pogonalin are given in Table 2.

Table 2: *Effect of pogonalin on the EC_{50} of [Pyr^1]apelin-13 induced β-arrestin recruitment*

[Pogonalin] (M)	EC_{50} of [[Pyr^1]apelin-13] (M)
0.00000005	0.00000011
0.00000015	0.00000028
0.0000004	0.00000072
0.000002	0.0000043
0.000006	0.000012

E) What is the affinity constant of the apelin receptor for pogonalin (4 marks)

In a subsequent experiment, the effects of pogonalin are tested on the ability of ELA-32 to recruit β-arrestin. Pogonalin was found to have no effect on the ability of ELA-32 to induce β-arrestin recruitment.

F) What can you conclude from these experiments and what further experiments could be carried out to confirm your conclusions? (3 marks)

End of Question 1

[Question 2 starts on the NEXT PAGE]

Question 2

Caffeine is one of the most frequently ingested drugs in the world. It is a naturally-occurring stimulant found in coffee, tea and chocolate.

Two studies were undertaken in order to better understand the pharmacokinetics of caffeine. In the first study, two groups of healthy male volunteers were recruited. One group were regular smokers. The other group had never smoked. The volunteers were asked not to consume caffeine-containing food or drinks for 36 hours prior to the study. A single oral dose of 100 mg caffeine was given. The bioavailability of caffeine in this formulation has previously been shown to be essentially complete (i.e. = 1) and is unaffected by smoking. Blood samples were taken at regular intervals. The plasma concentration of caffeine in these samples is shown in Table 1. The urine voided in the period between 2 to 3 hours was also collected. The total excretion of unchanged caffeine in urine in this period in the two groups is shown in Table 2.

Table 1: *Plasma concentration of caffeine in smokers and non-smokers. Data are the mean values in each group of volunteers.*

Time (Hours)	Caffeine concentration (ng/ml)	
	Smokers	Non-smokers
1	1711	1807
2	1307	1595
3	1096	1345
4	865	1134
6	577	846
10	269	461
24	14	58

Table 2: *Total amount of caffeine collected in urine voided between 2 to 3 hours. Data are the mean values for each group of volunteers.*

	Caffeine in urine (μg)
Smokers	108
Non-smokers	229

A) Calculate the terminal phase half-life, total clearance, and terminal phase volume of distribution of caffeine in the two groups of volunteers. (6 marks)

B) How is caffeine eliminated from the plasma? How might smoking affect elimination of caffeine? (3 marks)

C) Assuming that a cup of coffee contains 100 mg caffeine, what would be the average steady state plasma concentration of caffeine in the two groups of volunteers if they were to regularly drink 5 cups of coffee per day? Clearly state the assumptions that you have made in these calculations. (5 marks)

D) Briefly describe the possible consequences of quitting smoking on the effects of coffee. (2 marks)

[Question 2 continues on the NEXT PAGE]

In the second study, a further two groups of volunteers were recruited. The first group were obese (body mass index, BMI > 30), whereas the second group were of healthy weight (BMI 20 - 25). Both groups were non-smokers. The volunteers were asked to avoid caffeine-containing foods and drinks for 36 hours prior to the study. A single oral dose of 162 mg caffeine was given. Blood samples were regularly taken and used to calculate the pharmacokinetic values in Table 3.

Table 3: *Subject characteristics and pharmacokinetic values from the second study. Data are the mean values for each group of volunteers.*

	Obese (BMI > 30)	Healthy weight (BMI 20 – 25)
Weight (kg)	109	65
Terminal phase half-life (hours)	6.51	4.12
Area under curve, AUC (µg.hr/ml)	21.77	22.13

E) Using the data in Table 3, explain, as fully as possible, the effect of obesity on the pharmacokinetics of caffeine. (4 marks)

End of Question 2

[Question 3 starts on the NEXT PAGE]

Question 3

The combination of using genetically engineered knock-in animals and drugs has proven indispensable for understanding modes of drug action and roles of receptors and receptor subtypes in pathological conditions. A knock-in animal is one in which the gene of interest is replaced by an alternative version of the gene, for example one containing a mutation that changes the behaviour of the encoded protein.

In this study several different knock-in mice strains were generated containing a particular histidine (H) to arginine (R) mutation in either the α1, α2, α3 or α5 subunit of the $GABA_A$ receptor. This mutation blocks benzodiazepine binding.

These animals, alongside wild-type (WT) animals, were treated surgically to sensitize one paw (injured) to pain. They were then injected with vehicle (solution used to dissolve the drug, but without any drug added to it = negative control) or the $GABA_A$ receptor positive allosteric modulator, diazepam (D). Drugs were injected directly into the cerebrospinal fluid surrounding the spinal cord to avoid effects in the wider central nervous system. Injured and non-injured paw withdrawal latency in seconds (s), i.e. the time it took for the animal to withdraw the paw from a moderately painful heat stimulus, was measured as an indicator of sensitivity to pain, every 12 minutes. Increased sensitivity to pain leads to faster withdrawal.

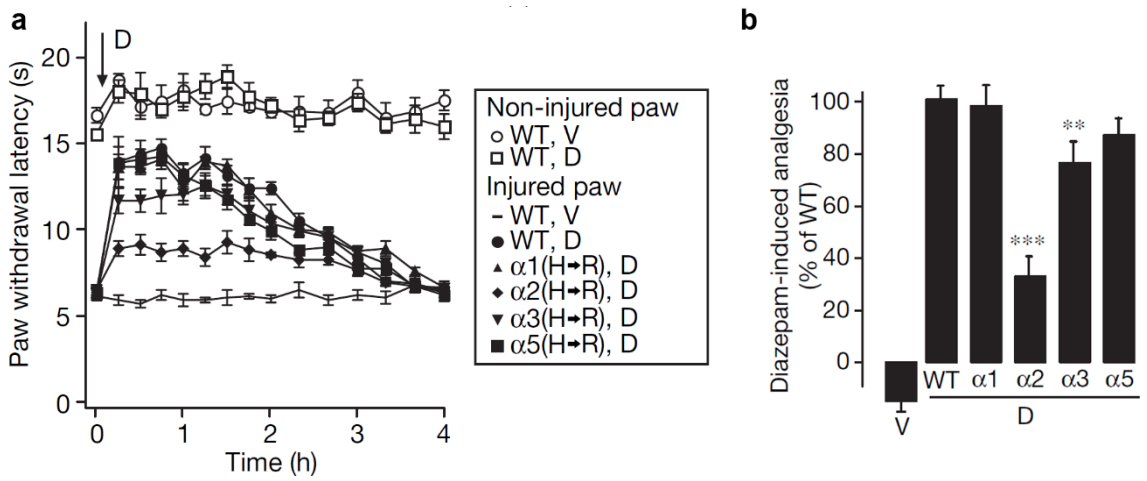

Figure 1: *(a) Sensitivity to pain of wild-type (WT) non-injured and injured paws after injection of vehicle or diazepam (D), and sensitivity to pain of knock-in α1, α2, α3 or α5 animals after injection of diazepam (0.1 mg/kg), versus time. (b) Summarized data from (a) measured as area under the curve (AUC) to give an overall measure of effect over the whole time course to gauge significance of effect. **, P<0.01; ***, P<0.001 (statistically significant against wild type; ANOVA followed by Bonferroni post-hoc test, n = 6 or 7 mice per group)*

A) What effect does diazepam have on paw withdrawal latency (pain threshold) in wild-type and knock-in animals? Which subtype(s) are important for mediating the effect of diazepam? Why were α4 and α6 knock-in animals not generated or tested? (4 marks)

[Question 3 continues on the NEXT PAGE]

Next, using the same experimental set-up as described earlier, the effect of an experimental drug, L-838,417 (L) with preferential positive allosteric modulation at the $\alpha 2$ subtype was investigated in wild-type mice.

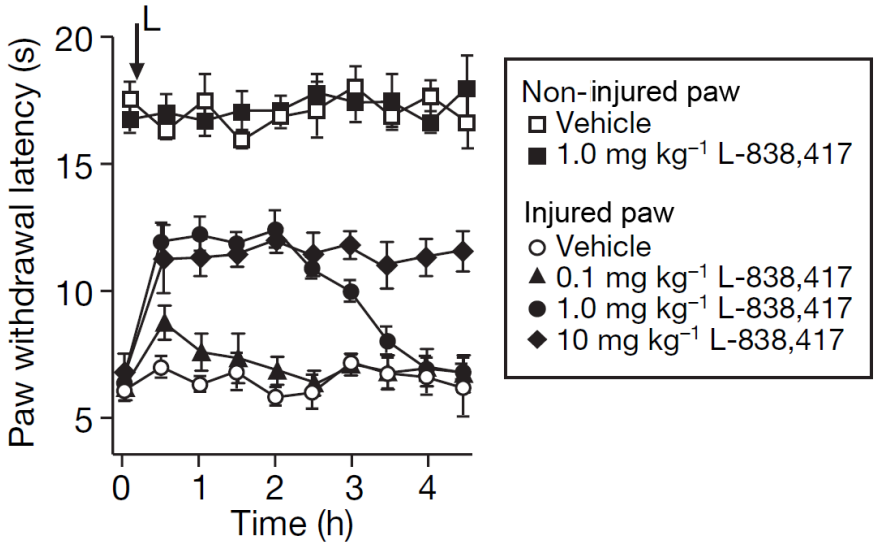

Figure 2: *Sensitivity to pain of non-injured and injured paws of wild-type animals after injection of vehicle or different doses of the experimental drug, L-838,417 (L), versus time.*

B) Do these data support or contradict the results in Figure 1? Explain why. (2 marks)

Next, using the same experimental set-up as described earlier, but featuring first an injection of L838,417 (L), then subsequent injection of either flumazenil (F) or naltrexone (N), the impacts of these drugs on the effect of L838,417 (L) was examined.

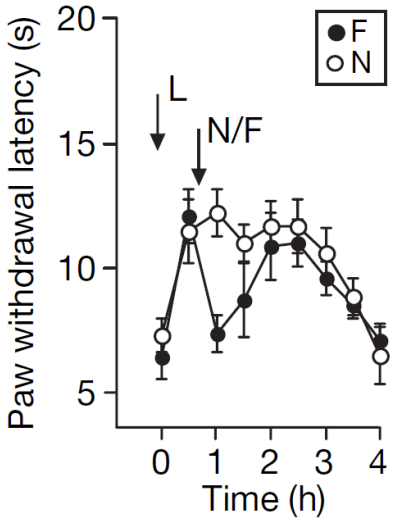

Figure 3: *Sensitivity to pain of the inflamed paw of wild-type animals after injection of the experimental drug, L-838,417 (L) at 1 mg/kg, and subsequent injection of either flumazenil (F) or naltrexone (N), each at 10 mg/kg, versus time.*

C) Explain the actions of these drugs. (3 marks)

[Question 3 continues on the NEXT PAGE]

Next, the effect of L838,417 (L) on performance in a rotarod test to assess general motor coordination was carried out.

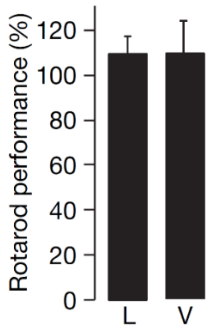

Figure 4: *Effects of L-838,417 (L; 1 mg/kg) versus vehicle (V) on motor coordination, shown as percentages of pre-drug rotarod performance.*

D) Why was this control experiment necessary? (1 mark)

In a follow-up study a reverse experiment to that done in Figure 1 was performed. Instead of studying knock-in mice strains with a single histidine (H) to arginine (R) mutation in either the $\alpha1$, $\alpha2$, $\alpha3$ or $\alpha5$ subunit of the $GABA_A$ receptor, the mutation was introduced into three of the four, or all four, subtypes in a single strain (subtype retaining a Histidine (H) is indicated in the graph legend by position of the H in the sequence of Rs).

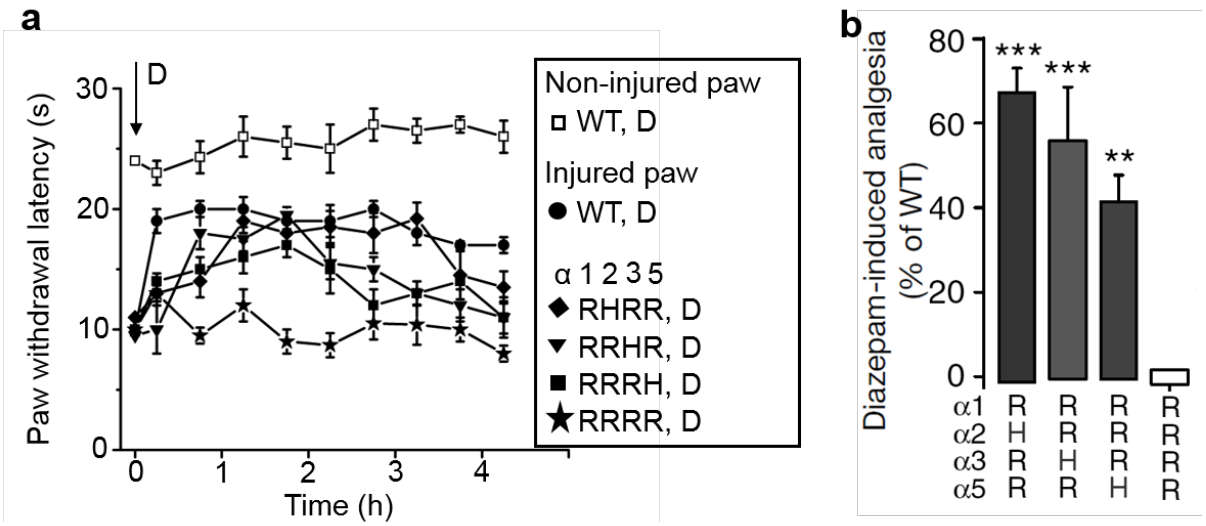

Figure 5: *Sensitivity to pain of wild-type (WT) non-injured and injured paws after injection of diazepam (D), and sensitivity to pain of combined knock-ins for $\alpha1$, $\alpha2$, $\alpha3$ or $\alpha5$ animals after injection of diazepam (0.1 mg/kg), versus time. Note: An animal with a histidine in only the $\alpha1$ subtype (i.e. HRRR) was not tested in this experiment. (b) Summarized data from (a) measured as area under the curve (AUC) to give an overall measure of effect over the whole time course to gauge significance of effect. Two asterisks, P<0.01; three asterisks, P<0.001 (statistically significant against wild type; ANOVA followed by Bonferroni post-hoc test, n=6 or 7 mice per group)*

E) What additional information is revealed from this study? How can differences be reconciled with Figure 1? What implications does this new data have for drug design compared to the old data? (4 marks)

[Question 3 continues on the NEXT PAGE]

Next, the effects of two drugs, diazepam (DZP) and midazolam (MDZ) were compared in cells expressing either $\alpha1$ or $\alpha2$-containing $GABA_A$ receptors. Increasing doses of these drugs were then administered orally to triple histidine (H) to arginine (R) mutation mouse strains, which retained only a single H in either $\alpha1$ or $\alpha2$. The effect on sedation for the $\alpha1$ HRRR animals, or on pain for the $\alpha2$ RHRR animals was measured. The drugs were radiolabelled to allow the receptor occupancy in the animals to also be measured in these experiments. The behavioural data were then plotted against receptor occupancy. Note, that sedation was tested but no effect observed on the $\alpha2$ RHRR animals (data not shown).

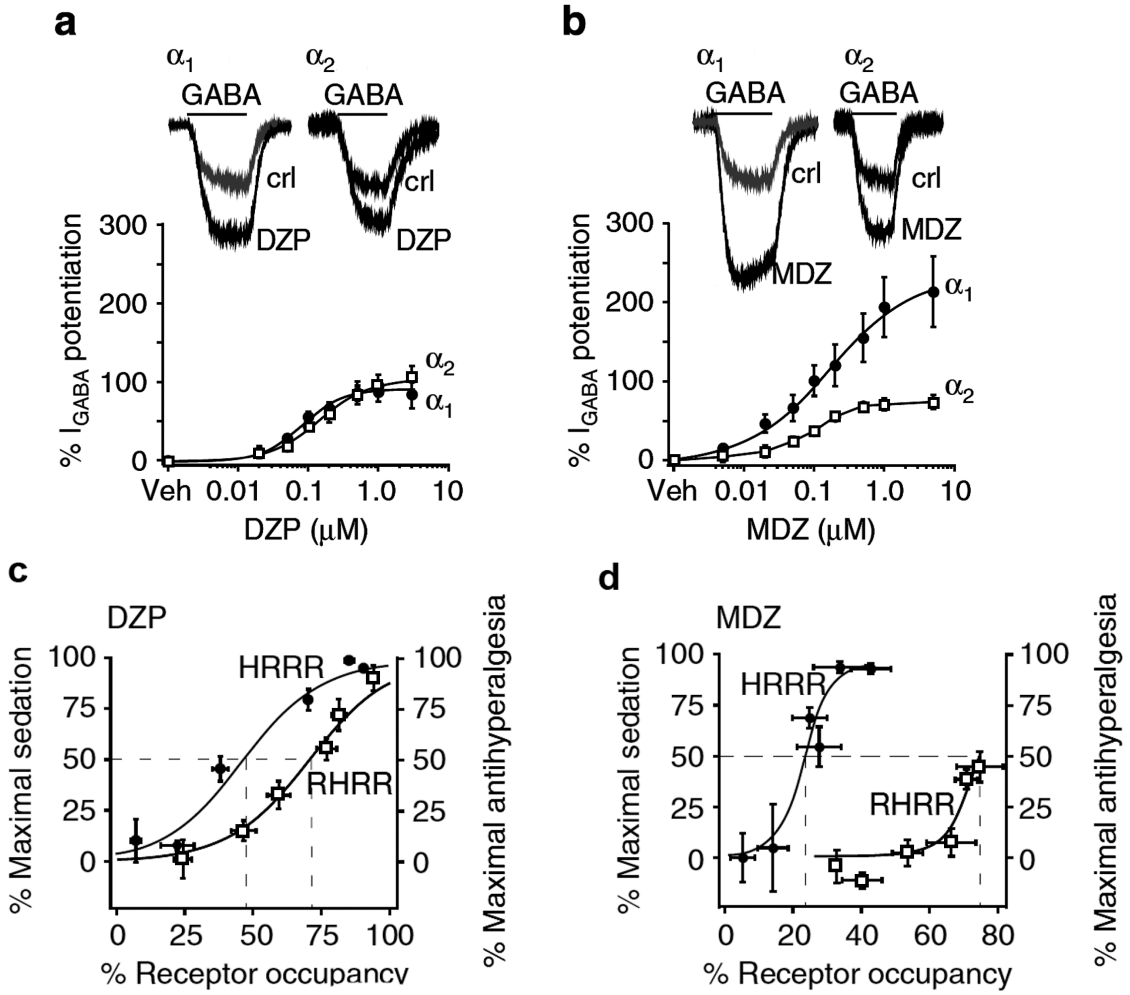

Figure 6: *(a) and (b) upper panels: overlays of current (I) recordings evoked by a low dose of GABA (GABA application duration indicated by black line) in the absence (control – crl) or presence of diazepam (DZP) or midazolam (MDZ). (a) and (b) lower panels: dose response curves for modulation by diazepam (DZP) or midazolam (MDZ) of GABA responses. (c) and (d) Percentages of maximal sedation measured in HRRR mice (retaining histidine only in $\alpha1$) and maximal pain relief (antihyperalgesia) measured in RHRR mice (retaining histidine only in $\alpha2$) plotted against receptor occupancy.*

F) Explain the effects of these drugs. Based on this data are these drugs suitable for the treatment of pain disorders? Are $GABA_A$ receptors good targets for the development of pain therapeutics? (6 marks)

END OF PAPER